Supplementary Material for *Signer-disjoint evaluation and Russian-to-Uzbek transfer for isolated sign recognition: a multi-signer Uzbek Sign Language corpus and a compact pose encoder*

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All numbers were computed from the per-recording predictions of the runs by the analysis script released with the code (`tools/analyze.py`); confidence intervals are percentile bootstrap intervals with 2,000 resamples over test signers (SD-5) or 10,000 resamples over signers (LOSO).

S1 Full ablation sweep on Slovo (SD-5)

Table S1 lists every input, architecture, training and capacity variant trained on Slovo under SD-5 with one seed and compared with MPE-A0 (no weight averaging) by the paired difference over test signers; the five main ablations of MPE-A1 with three seeds are in Fig. 5 of the article. The last column gives the number of test signers (of 188) for which the variant is more accurate than MPE-A0.

Table S1: Variants of MPE-A0 on Slovo under SD-5 (one seed, five folds): top-1 accuracy [95% CI over test signers], paired difference Δ to MPE-A0 [95% CI] and the number of test signers with a positive difference.

Variant	**Top-1 [95% CI]**	**Δ vs MPE-A0 [95% CI]**	**Signers +**
Reference: MPE-A0 (69.3 [67.2; 71.2], three seeds)			
Input and normalisation			
No normalisation (raw coordinates)	64.3 [61.8; 66.5]	−5.0 [−6.3; −3.7]	52 / 188
Per-frame instead of per-clip normalisation	68.7 [66.4; 70.8]	−0.6 [−1.3; +0.2]	78 / 188
Hands only (no body, no face points)	60.9 [58.8; 62.8]	−8.4 [−9.6; −7.2]	34 / 188
No body points	66.8 [64.6; 68.9]	−2.5 [−3.5; −1.7]	60 / 188
No face points	68.9 [66.7; 71.0]	−0.5 [−1.1; +0.3]	86 / 188
No mask channel	69.5 [67.3; 71.5]	+0.2 [−0.6; +1.0]	78 / 188
Depth z added	69.6 [67.4; 71.7]	+0.3 [−0.6; +1.2]	83 / 188
Joint angles added	69.5 [67.6; 71.4]	+0.2 [−0.6; +0.9]	83 / 188
Motion (frame-difference) features added	70.1 [68.0; 72.1]	+0.8 [−0.0; +1.6]	92 / 188
Dominant-hand mirroring	68.6 [66.4; 70.6]	−0.7 [−1.6; +0.1]	74 / 188
Spatial encoder			
No graph: per-part MLP (as in MPE-A2)	68.4 [66.3; 70.6]	−0.9 [−1.8; +0.0]	73 / 188
No long-range edges	67.2 [64.8; 69.3]	−2.1 [−3.1; −1.1]	64 / 188
No part tokens: one token per frame	67.3 [65.0; 69.5]	−2.0 [−2.9; −1.0]	67 / 188
No bone stream	68.5 [66.2; 70.6]	−0.8 [−1.6; −0.0]	74 / 188
Alternative set of hand nodes	68.8 [66.5; 71.0]	−0.5 [−1.3; +0.2]	77 / 188
Shared weights for the two hands	68.9 [66.8; 70.9]	−0.4 [−1.2; +0.3]	79 / 188
Mask-aware batch normalisation	69.0 [67.0; 70.9]	−0.3 [−0.9; +0.4]	77 / 188
Late fusion of joint and bone streams	70.3 [68.4; 72.1]	+1.0 [+0.3; +1.8]	87 / 188
Temporal encoder and pooling			
No depthwise temporal convolution	68.3 [66.3; 70.1]	−1.0 [−1.8; −0.2]	71 / 188
Transformer layers in place of Conformer layers	67.8 [65.8; 69.8]	−1.5 [−2.2; −0.6]	81 / 188
Mean pooling in place of attention pooling	67.1 [64.8; 69.3]	−2.2 [−3.3; −1.2]	61 / 188
Training			
No augmentation	68.5 [66.5; 70.4]	−0.8 [−1.8; +0.1]	72 / 188
No auxiliary heads	68.5 [66.5; 70.5]	−0.8 [−1.6; +0.1]	75 / 188
Capacity and window			
$D = 128$	68.8 [66.6; 70.9]	−0.5 [−1.3; +0.3]	82 / 188
$D = 256$	69.3 [67.3; 71.3]	+0.0 [−0.7; +0.8]	83 / 188
$N = 2$	68.8 [66.7; 71.0]	−0.6 [−1.2; +0.2]	74 / 188
$N = 4$	69.9 [67.7; 72.1]	+0.6 [−0.2; +1.4]	93 / 188
$T = 48$	68.7 [66.5; 70.8]	−0.6 [−1.6; +0.3]	75 / 188
$D = 256$, $N = 6$	72.0 [69.8; 73.9]	+2.6 [+2.0; +3.3]	114 / 188
$D = 384$, $N = 8$	72.1 [70.2; 74.0]	+2.8 [+1.9; +3.7]	109 / 188

S2 Accuracy per held-out signer on the UzSL corpus

Table S2 gives the top-1 accuracy of every condition of Table 5 of the article on each held-out signer (mean of three seeds), the number of evaluation-set recordings per signer, and the accuracy on words and on fingerspelling for the main model. S06 and S10 (fewer than 20 recordings) form the "other" fold and are excluded from the macro-average.

Table S2: LOSO on the UzSL corpus: top-1 accuracy (%) per held-out signer, mean of three seeds; last column: macro-average over the ten signers.

Condition	**S01**	**S02**	**S03**	**S04**	**S05**	**S07**	**S08**	**S09**	**S11**	**S12**	**Macro**
Recordings (evaluation set)	382	1,095	590	125	183	648	71	74	118	118	3,404
MPE-A1, pretrained (Slovo + Bukva), 682-class head	24.9	21.0	17.2	18.1	31.3	27.4	42.7	40.5	51.1	45.5	32.0
MPE-A1, from scratch, 682-class head	11.9	8.7	9.1	9.9	15.7	15.5	16.0	16.7	34.5	35.3	17.3
SL-GCN (released), pretrained, 682-class head	24.8	22.6	21.0	18.9	33.5	29.4	49.3	42.3	53.4	53.7	34.9
MPE-A1, pretrained (Slovo + Bukva + ASL), 682-class head	25.0	21.7	19.5	22.4	34.2	28.8	40.4	38.3	39.3	33.9	30.3
MPE-A1, pretrained, all recordings (2,033–2,378-class head)	19.8	16.8	13.1	14.9	27.1	22.7	40.9	29.3	44.1	33.0	26.2
MPE-A1, from scratch, all recordings	6.7	5.7	5.7	7.7	12.6	12.4	11.7	8.6	31.9	37.0	14.0
Top-1 on words / fingerspelling, MPE-A1 pretrained (signers with at least ten recordings of the group)											
Words	24.9	21.0	17.5	18.1	31.5	27.8	42.7	40.5	—	—	28.0
Fingerspelling	—	—	8.3	—	—	—	—	—	51.1	45.5	35.0

S3 Per-class precision, recall and F_1

Tables S3 and S4 give, for every class, the number of test recordings per seed (support), precision, recall and F_1 averaged over seeds: Bukva under SD-5 (MPE-A1, pooled over the five folds), the UzSL pilot (MPE-A1 pretrained on Slovo + Bukva and fine-tuned on all recordings, pooled over the two test participants) and the 682-class evaluation set of the UzSL corpus under LOSO (MPE-A1 pretrained on Slovo + Bukva, pooled over the ten folds). Pilot classes with zero support in the training data of their fold have precision and recall equal to zero. The same tables are provided as CSV files with the code.

Table S3: Per-class metrics (%) on Bukva (left; 33 letters of the RSL manual alphabet, SD-5) and on the UzSL pilot (right; 29 letters and 11 numerals). n: test recordings per seed.

Class	n	**P**	**R**	F_1	**Class**	n	**P**	**R**	F_1
Ё	123	97.0	95.9	96.5	0	4	0.0	0.0	0.0
А	118	99.4	99.2	99.3	1	6	0.0	0.0	0.0
Б	109	95.5	97.9	96.7	10	5	0.0	0.0	0.0
В	116	98.6	99.4	99.0	2	5	0.0	0.0	0.0
Г	112	97.6	97.3	97.5	3	5	0.0	0.0	0.0
Д	102	92.9	94.1	93.5	4	5	0.0	0.0	0.0
Е	119	89.5	93.0	91.2	5	6	0.0	0.0	0.0
Ж	118	93.3	89.5	91.4	6	5	0.0	0.0	0.0
З	108	97.2	97.2	97.2	7	5	33.3	33.3	33.3
И	118	95.1	92.9	94.0	8	7	0.0	0.0	0.0
Й	107	98.8	99.7	99.2	9	5	0.0	0.0	0.0
К	107	96.0	90.0	92.9	A	10	75.1	100.0	85.7
Л	117	82.7	84.0	83.3	B	10	23.8	16.7	19.6
М	104	83.8	82.4	83.0	CH	10	66.7	33.3	37.2
Н	118	98.3	95.8	97.0	D	10	40.0	80.0	53.3
О	120	97.5	98.1	97.8	E	10	0.0	0.0	0.0
П	118	85.2	86.2	85.7	F	10	97.0	100.0	98.4
Р	115	98.5	96.2	97.4	G	10	46.1	80.0	57.6
С	125	100.0	97.9	98.9	G'	10	31.8	20.0	21.4
Т	114	82.7	84.2	83.5	H	5	0.0	0.0	0.0
У	113	98.2	97.9	98.1	I	5	0.0	0.0	0.0
Ф	116	96.0	97.1	96.6	J	9	50.9	100.0	65.6
Х	121	95.4	96.7	96.0	K	10	63.9	76.7	69.7
Ц	112	98.5	98.5	98.5	L	11	0.0	0.0	0.0
Ч	107	97.2	99.1	98.1	M	10	64.1	86.7	69.7
Ш	119	68.2	75.9	71.8	N	10	80.8	46.7	58.5
Щ	103	70.6	61.5	65.6	NG	9	89.2	81.5	84.7
Ъ	115	96.2	96.5	96.4	O	10	65.0	93.3	76.5
Ы	123	99.2	100.0	99.6	O'	10	73.8	30.0	40.2
Ь	101	94.7	95.0	94.9	P	9	47.3	44.4	44.7
Э	110	95.4	93.9	94.7	Q	5	0.0	0.0	0.0
Ю	117	94.8	98.3	96.5	R	10	57.8	90.0	66.4
Я	112	90.2	90.2	90.2	S	10	93.9	100.0	96.8
					SH	10	65.1	96.7	76.0
					T	8	55.1	25.0	31.2
					U	10	53.8	93.3	68.3
					V	5	0.0	0.0	0.0
					X	10	32.6	50.0	39.3
					Y	5	0.0	0.0	0.0
					Z	10	55.6	83.3	66.6

Table S4: Per-class metrics (%) on the 682-class evaluation set of the UzSL corpus (LOSO, MPE-A1 pretrained on Slovo + Bukva, 682 classes in alphabetical order; "harf:" marks fingerspelled letters, "raqam:" numerals). n: test recordings per seed.

Class	n	P	R	F_1	Class	n	P	R	F_1	Class	n	P	R	F_1
abadiy	6	33.3	27.8	30.3	ishtaha	4	0.0	0.0	0.0	qo'y	9	53.2	55.6	54.0
abort	3	0.0	0.0	0.0	ism	5	52.8	26.7	34.8	qog'oz	4	33.3	8.3	13.3
adabiyot	3	33.3	11.1	16.7	ispaniya	4	15.0	16.7	15.7	qoida	4	73.9	91.7	81.3
advokat	4	18.7	25.0	21.1	issiq	6	0.0	0.0	0.0	qon ketish	5	4.2	6.7	5.1
afrika	3	0.0	0.0	0.0	it	3	0.0	0.0	0.0	qonun	5	69.0	66.7	66.7
aka	4	0.0	0.0	0.0	italiya	3	33.3	22.2	26.7	qor	6	17.6	16.7	16.2
akrobatika	6	0.0	0.0	0.0	iyul	3	0.0	0.0	0.0	qora	6	11.1	16.7	13.3
aktyor	3	46.7	55.6	50.0	iyun	7	30.3	38.1	33.7	qoshiq	9	34.8	51.9	41.0
albom	3	17.3	33.3	22.3	jadval	5	15.5	26.7	19.5	qovun	4	31.1	33.3	31.7
amaki	4	36.8	75.0	49.1	jarima	7	37.0	42.9	39.4	qovurmoq	4	34.0	41.7	36.5
analiz	4	0.0	0.0	0.0	jazolamoq	4	0.0	0.0	0.0	qozon	11	15.7	18.2	16.5
anar	3	0.0	0.0	0.0	jigarrang	4	33.9	50.0	40.0	qulf	3	0.0	0.0	0.0
angina	3	0.0	0.0	0.0	jirafa	5	33.3	33.3	32.5	quloqchin	4	59.3	75.0	65.4
angliya	3	22.2	22.2	20.7	jismoniy tarbiya	5	0.0	0.0	0.0	qulupnay	3	0.0	0.0	0.0
aniq	10	42.2	36.7	38.8	jiyan	5	0.0	0.0	0.0	qum	4	0.0	0.0	0.0
apelsin	4	0.0	0.0	0.0	juma	6	7.9	11.1	9.2	qurbaqa	7	15.2	14.3	13.7
aprel	5	39.7	46.7	41.7	kabob	4	57.4	50.0	49.1	qurt	6	41.7	11.1	16.2
archa	5	24.4	46.7	32.1	kafel	5	0.0	0.0	0.0	qush	4	0.0	0.0	0.0
aroq	5	33.3	6.7	11.1	kalbasa	4	0.0	0.0	0.0	quymoq	4	0.0	0.0	0.0
arzon	5	30.0	20.0	22.9	kalendar	3	0.0	0.0	0.0	quyon	5	8.3	6.7	7.4
asalari	3	0.0	0.0	0.0	kalish	4	45.6	41.7	43.1	quyosh	8	44.3	58.3	49.7
avgust	3	33.3	11.1	16.7	kalit	4	8.3	8.3	8.3	radio	4	6.7	8.3	7.4
avlod	3	0.0	0.0	0.0	kam	3	0.0	0.0	0.0	rahbar	5	20.0	20.0	20.0
avtobus	10	43.9	66.7	52.8	kamalak	6	0.0	0.0	0.0	rahmat	6	25.8	55.6	34.9
avtomobil	4	0.0	0.0	0.0	karam	4	0.0	0.0	0.0	rais	5	27.6	40.0	32.6
ayiq	6	44.4	27.8	32.8	karate	5	0.0	0.0	0.0	rasm	8	19.6	25.0	21.7
ayol	6	24.9	50.0	33.2	karidor	5	30.6	20.0	24.1	rejissyor	3	57.8	66.7	61.1
aytmoq	4	26.4	33.3	29.2	karta	4	0.0	0.0	0.0	restoran	3	83.3	100.0	88.9
ayyor	3	16.5	33.3	19.7	kartoshka	7	0.0	0.0	0.0	riga	3	0.0	0.0	0.0
ba'zan	7	45.6	90.5	60.6	katta	8	11.1	4.2	6.1	ritmika	3	25.0	33.3	28.0
baho	5	0.0	0.0	0.0	kayfiyat	4	0.0	0.0	0.0	ro'mol	5	27.0	26.7	26.0
bahor	8	45.2	75.0	56.4	kech	3	0.0	0.0	0.0	rossiya	5	49.0	66.7	56.1
baliq	7	0.0	0.0	0.0	kecha	6	61.3	83.3	70.3	ruchka	7	27.7	61.9	37.2
balkon	5	0.0	0.0	0.0	kekkaygan	5	16.7	13.3	14.8	sabzavotlar	4	29.8	50.0	36.6
banan	3	0.0	0.0	0.0	kenguru	7	46.2	52.4	49.0	sabzi	5	0.0	0.0	0.0
bank	6	29.1	50.0	36.8	kepka	6	0.0	0.0	0.0	sakramoq	5	0.0	0.0	0.0
banka	8	45.2	41.7	42.1	kerak	5	24.3	33.3	28.0	sakson	4	0.0	0.0	0.0
bantik	4	0.0	0.0	0.0	keyin	4	0.0	0.0	0.0	salom	6	11.1	5.6	7.4
basketbol	6	0.0	0.0	0.0	kichik	8	3.0	4.2	3.5	samalyot	3	0.0	0.0	0.0
basseyn	4	0.0	0.0	0.0	kim	4	39.7	41.7	39.4	samolyot	3	0.0	0.0	0.0
baxtsiz	3	11.1	11.1	11.1	kimyo	3	0.0	0.0	0.0	sana	4	73.3	33.3	43.0
begemot	3	16.7	11.1	13.3	kimyoviy tozalash	5	0.0	0.0	0.0	sanchqi	7	36.6	38.1	36.8
bekat	4	41.7	83.3	54.7	kino	3	0.0	0.0	0.0	sarafan	3	33.3	33.3	32.4
belarus	4	25.0	16.7	19.4	kiryuvish mashinasi	4	33.7	41.7	36.0	sarimsoq	3	0.0	0.0	0.0
belbog	3	0.0	0.0	0.0	kiyik	5	0.0	0.0	0.0	sariq	6	49.2	50.0	49.1
bermayman	3	0.0	0.0	0.0	kiyinmoq	5	0.0	0.0	0.0	saryog	3	0.0	0.0	0.0
besh	5	25.0	26.7	25.6	ko'cha	4	72.5	91.7	79.6	saxiy	4	0.0	0.0	0.0
bilmoq	5	18.6	33.3	23.6	ko'k	8	29.6	33.3	30.7	sekin	6	0.0	0.0	0.0
bir kun	4	33.3	16.7	22.2	ko'l	4	6.7	8.3	7.4	sen	3	0.0	0.0	0.0
bir yildan keyin	3	0.0	0.0	0.0	ko'p	3	46.7	55.6	50.0	senda	2	0.0	0.0	0.0
bit	3	33.3	11.1	16.7	ko'prik	6	54.1	55.6	53.2	sentyabr	5	21.4	20.0	20.4
biz	4	27.8	25.0	26.2	ko'rpa	6	41.1	55.6	45.5	seshanba	6	11.1	5.6	7.4
blinchik	5	0.0	0.0	0.0	ko'ylak	6	36.5	66.7	46.8	shafqatsiz	3	0.0	0.0	0.0
bo'ladi	4	0.0	0.0	0.0	ko'zgu	3	0.0	0.0	0.0	shaftoli	5	6.7	6.7	6.7
bo'ri	4	47.6	75.0	58.2	ko'zoynak	3	0.0	0.0	0.0	shahar	3	0.0	0.0	0.0
bo'tqa	5	0.0	0.0	0.0	kofe	3	0.0	0.0	0.0	shakar	6	47.2	44.4	44.4
bo'ydoq	3	66.7	44.4	50.0	kofta	6	25.2	33.3	28.3	shalgam	4	33.3	8.3	13.3
bobo	8	21.1	25.0	22.9	kolgotka	3	0.0	0.0	0.0	shamol	4	29.4	50.0	36.0
boks	5	29.4	46.7	35.7	konfet	4	0.0	0.0	0.0	shampan vinosi	4	16.7	16.7	16.7
bola	6	25.0	22.2	22.5	konki	5	0.0	0.0	0.0	shanba	6	8.4	16.7	11.2
bolalar	3	0.0	0.0	0.0	konserva	9	37.9	33.3	35.4	shapka	4	67.9	100.0	80.5
boshlamoq	3	33.3	11.1	16.7	koreya	3	52.8	44.4	47.3	sharf	5	61.0	66.7	63.3
botinka	4	45.1	50.0	46.0	kostyum	9	22.8	22.2	21.3	sharq	3	0.0	0.0	0.0
boy	5	42.2	33.3	35.9	koza	3	0.0	0.0	0.0	shaxmat	6	39.5	33.3	35.7
bozor	5	46.7	33.3	38.3	kredit	5	0.0	0.0	0.0	sher	9	64.3	59.3	60.7
bug'doy yormasi	4	0.0	0.0	0.0	kreslo	11	69.5	60.6	64.5	shifokor	5	22.6	33.3	26.9
bugun	8	58.1	54.2	54.7	krujka	4	0.0	0.0	0.0	shift	3	0.0	0.0	0.0
buka	6	15.9	11.1	12.5	kuchli	5	0.0	0.0	0.0	shim	7	0.0	0.0	0.0
bulut	4	44.4	25.0	31.7	kulrang	5	39.5	46.7	42.6	shippak	8	10.1	12.5	11.1
butilka	6	0.0	0.0	0.0	kun	5	27.0	20.0	22.2	shirin	7	4.8	4.8	4.8
buvi	3	0.0	0.0	0.0	kurash	7	15.9	14.3	15.0	shkaf	3	0.0	0.0	0.0
buxoro	3	0.0	0.0	0.0	kurka	6	28.4	27.8	27.9	shlyapa	3	0.0	0.0	0.0
chang	4	0.0	0.0	0.0	kurtka	8	37.8	29.2	31.7	sho'rva	8	23.3	37.5	28.0
chang'i	7	18.7	19.0	18.8	kutubxona	4	38.9	25.0	30.2	shvetsiya	3	0.0	0.0	0.0
changyutgich	4	38.9	33.3	35.7	kuz	11	54.2	48.5	50.9	sibir	4	16.7	8.3	11.1
chaqmoq	5	31.7	33.3	32.0	kvartira	5	0.0	0.0	0.0	sichqon	5	20.7	40.0	27.3
chashka	4	53.3	33.3	35.6	lagan	3	0.0	0.0	0.0	sigir	7	95.2	85.7	90.1
chempion	4	61.1	25.0	34.0	laqab	4	36.1	25.0	29.0	sinf	3	0.0	0.0	0.0
chet el	3	0.0	0.0	0.0	latvia	3	0.0	0.0	0.0	siz	6	0.0	0.0	0.0
chigirtka	4	0.0	0.0	0.0	lift	6	54.2	66.7	59.7	so'z	3	46.7	55.6	49.0
chipta	4	0.0	0.0	0.0	likopcha	5	0.0	0.0	0.0	soat	11	3.5	6.1	4.4
chiroq	5	63.3	53.3	56.2	limon	6	41.3	55.6	47.2	sochiq	8	0.0	0.0	0.0
chiroyli	5	0.0	0.0	0.0	litva	4	56.1	100.0	71.4	somsa	4	33.3	8.3	13.3
chivin	3	0.0	0.0	0.0	loy	4	0.0	0.0	0.0	soniya	3	4.2	11.1	6.1
chizg'ich	4	8.3	8.3	8.3	magnitafon	3	35.6	44.4	38.9	sosiska	6	62.7	61.1	61.2
cho'chqa	5	54.8	66.7	59.0	majlis	4	44.4	41.7	42.4	sotib olmoq	3	0.0	0.0	0.0
cho'mich	4	4.2	8.3	5.6	makaron	7	28.8	33.3	30.1	sotmoq	3	0.0	0.0	0.0
chodir	4	8.3	8.3	8.3	maktab	3	0.0	0.0	0.0	sovuq	4	0.0	0.0	0.0
chorraha	6	24.6	38.9	30.1	mamont	3	0.0	0.0	0.0	sport	7	13.3	23.8	17.0
chorshanba	6	8.3	5.6	6.7	manti	4	5.6	8.3	6.7	stadion	5	0.0	0.0	0.0
choy	13	40.8	53.8	46.2	manzil	4	63.3	33.3	39.3	stakan	5	22.6	26.7	24.1
choynak	9	0.0	0.0	0.0	maosh	5	61.1	46.7	51.5	stol	7	74.6	52.4	59.6
choyshab	3	0.0	0.0	0.0	maqtamoq	5	6.7	6.7	6.7	stul	6	39.8	44.4	41.1
chuchvara	5	0.0	0.0	0.0	maqtanchoq	3	16.7	11.1	13.3	sud	3	27.8	33.3	30.2
chumoli	6	0.0	0.0	0.0	markaz	4	13.3	16.7	14.8	supurgi	12	9.2	11.1	10.1
daftar	5	0.0	0.0	0.0	mart	8	17.8	37.5	24.1	sut	4	44.4	50.0	46.6
dam olmoq	4	8.3	8.3	8.3	masala	3	0.0	0.0	0.0	sutka	3	0.0	0.0	0.0
daniya	4	20.6	33.3	25.5	mashina	5	0.0	0.0	0.0	suv	5	6.7	6.7	6.7
daqiqa	6	0.0	0.0	0.0	mavzu	3	83.3	100.0	90.5	suyak	3	0.0	0.0	0.0
dars	6	6.7	11.1	8.3	may	6	14.1	27.8	18.7	suzish	3	38.1	22.2	23.3
darvoza	10	18.1	20.0	18.9	mayka	7	31.5	47.6	37.5	ta'til	3	11.1	11.1	11.1
darvozabon	3	61.7	100.0	75.8	maymun	8	29.4	29.2	29.2	tabiat	3	63.3	55.6	52.4
daryo	3	0.0	0.0	0.0	mazali	5	41.0	60.0	47.8	taklif	3	0.0	0.0	0.0
dastro'mol	4	41.1	75.0	53.0	mebel	5	0.0	0.0	0.0	taksi	7	27.3	28.6	27.6
dasturxon	8	35.4	37.5	36.2	mehmonxona	3	33.3	22.2	26.7	tana	3	68.9	66.7	65.6
davlat	3	66.7	88.9	75.0	men	5	0.0	0.0	0.0	tanaffus	5	64.9	66.7	62.6
dekabr	6	32.4	38.9	33.4	menda	3	0.0	0.0	0.0	tarelka	4	0.0	0.0	0.0
delfin	7	62.1	61.9	60.9	menga	2	16.7	16.7	16.7	tarix	3	66.7	66.7	64.8
dengiz	3	13.2	44.4	20.2	metro	7	46.7	33.3	38.9	tarjimon	4	0.0	0.0	0.0
deraza	5	52.5	33.3	33.2	mevalar	3	42.9	66.7	50.4	tartibli	4	0.0	0.0	0.0
devor	12	36.0	41.7	37.0	milliard	7	0.0	0.0	0.0	tarvuz	3	0.0	0.0	0.0
diqqat	3	0.0	0.0	0.0	milliy	3	0.0	0.0	0.0	tayyorlamoq	5	0.0	0.0	0.0
direktor	3	0.0	0.0	0.0	moldova	4	57.6	91.7	70.0	teatr	4	100.0	75.0	85.7
divan	7	0.0	0.0	0.0	moskva	3	0.0	0.0	0.0	tekshirmoq	4	0.0	0.0	0.0
do'kon	4	8.3	8.3	8.3	mototsikl	4	8.3	8.3	8.3	telefon	5	40.0	33.3	36.3

Table S4: (continued)

Class	n	P	R	F_1	Class	n	P	R	F_1	Class	n	P	R	F_1
do'l	3	0.0	0.0	0.0	mushuk	8	0.0	0.0	0.0	telegraf	5	10.8	13.3	11.8
do'st	4	0.0	0.0	0.0	musiqa	3	0.0	0.0	0.0	televizor	7	72.0	85.7	78.0
doim	8	0.0	0.0	0.0	musobaqa	4	0.0	0.0	0.0	temir yo'l	3	0.0	0.0	0.0
domino	6	19.4	11.1	14.1	muzlatgich	11	35.8	21.2	26.5	tennis	3	0.0	0.0	0.0
dori	3	6.7	11.1	8.3	muzqaymoq	3	0.0	0.0	0.0	terapevt	4	33.3	8.3	13.3
dorixona	7	0.0	0.0	0.0	nafratlanmoq	3	0.0	0.0	0.0	teri	4	6.7	8.3	7.4
doska	3	0.0	0.0	0.0	namangan	3	0.0	0.0	0.0	terma	3	0.0	0.0	0.0
dush	3	64.3	55.6	50.0	navbatchi	4	50.4	66.7	56.5	tez	9	0.0	0.0	0.0
dushanba	6	55.9	77.8	63.6	nima	6	49.2	44.4	45.8	tez-tez	3	0.0	0.0	0.0
echki	6	4.2	5.6	4.8	nok	3	48.9	66.7	55.6	tikuvchi	3	13.9	22.2	16.9
egizaklar	4	61.1	25.0	34.0	nol	3	0.0	0.0	0.0	timsoh	4	38.7	58.3	46.3
ekologiya	3	83.3	100.0	88.9	non	5	3.7	6.7	4.8	tipratikan	3	0.0	0.0	0.0
elak	6	42.5	38.9	39.7	novosibirsk	3	0.0	0.0	0.0	to'qimoq	3	0.0	0.0	0.0
elektir	4	0.0	0.0	0.0	noyabr	6	23.3	44.4	30.5	to'qqiz	5	55.0	53.3	53.7
erkak	8	34.5	33.3	33.9	o'g'lim	7	0.0	0.0	0.0	to'qson	3	0.0	0.0	0.0
erkaklar ko'ylagi	3	80.6	77.8	77.5	o'g'rilik	8	0.0	0.0	0.0	to'rt	7	13.1	9.5	10.8
ertaga	4	9.5	16.7	12.1	o'lim	6	53.5	77.8	62.1	to'satdan	3	0.0	0.0	0.0
ertage	3	0.0	0.0	0.0	o'n olti	3	61.7	88.9	72.6	to'y	7	39.2	57.1	45.7
ertak	4	86.7	100.0	92.6	o'n to'rt	4	0.0	0.0	0.0	tom	3	59.7	88.9	69.0
ertalab	3	0.0	0.0	0.0	o'n uch	3	0.0	0.0	0.0	tong	5	20.6	26.7	23.2
eshik	7	25.8	38.1	30.7	o'qimoq	5	11.1	13.3	11.3	topshiriq	4	0.0	0.0	0.0
eshitmoq	3	25.4	44.4	31.7	o'qlov	8	51.0	66.7	57.6	tova	5	23.3	26.7	23.4
eslamoq	5	39.3	60.0	47.4	o'quvchi	4	4.8	8.3	6.1	tovuq	6	58.3	83.3	68.5
estoniya	4	0.0	0.0	0.0	o'rdak	8	29.4	33.3	30.9	tozalamoq	7	72.2	14.3	21.8
etik	6	38.1	50.0	43.2	o'rganmoq	3	0.0	0.0	0.0	tramvay	3	0.0	0.0	0.0
fabrika	3	0.0	0.0	0.0	o'rgimchak	7	13.9	19.0	15.9	trolleybus	5	0.0	0.0	0.0
faks	6	17.8	11.1	13.5	o'rik	4	11.1	8.3	9.5	tufli	6	0.0	0.0	0.0
falaj	4	16.7	8.3	11.1	o'rmon	3	33.3	11.1	16.7	tug'ilmoq	3	0.0	0.0	0.0
familiya	4	0.0	0.0	0.0	o'simlik yog'i	4	0.0	0.0	0.0	tulki	5	0.0	0.0	0.0
farrosh	8	0.0	0.0	0.0	o'smir	3	0.0	0.0	0.0	tumov	4	83.3	33.3	43.3
fevral	5	0.0	0.0	0.0	o'tgan kuni	3	52.8	77.8	62.4	tun	3	80.0	66.7	70.0
figurali uchish	6	0.0	0.0	0.0	o'tgan yil	4	42.9	50.0	45.5	turmoq	5	43.5	73.3	54.3
fikr	3	0.0	0.0	0.0	o'yin	6	49.0	72.2	58.3	tushlik	3	72.2	55.6	62.2
fil	7	27.2	38.1	31.2	o'z	2	0.0	0.0	0.0	tushunmaslik	4	59.3	75.0	65.4
finlandiya	3	0.0	0.0	0.0	o'zbekiston	4	0.0	0.0	0.0	tushunmoq	4	0.0	0.0	0.0
foyda	3	0.0	0.0	0.0	o'zimniki	3	33.3	66.7	44.4	tuxum	10	20.2	16.7	17.9
futbol	6	35.4	27.8	29.5	oddiy	4	0.0	0.0	0.0	tuya	4	0.0	0.0	0.0
futbolka	7	39.8	66.7	49.7	odessa	3	0.0	0.0	0.0	tuz	7	10.8	23.8	14.9
g'alaba	5	57.8	40.0	44.5	oila	6	6.7	5.6	6.1	tvorog	4	61.1	41.7	49.2
g'amgin	4	44.4	41.7	42.4	oktabr	4	55.6	25.0	32.4	uch	5	58.3	26.7	33.3
g'arb	3	33.3	11.1	16.7	oldin	3	0.0	0.0	0.0	uka	3	0.0	0.0	0.0
g'oz	4	0.0	0.0	0.0	olimpiada	6	41.2	66.7	50.3	ukraina	3	0.0	0.0	0.0
galstuk	4	70.0	58.3	61.1	olma	9	0.0	0.0	0.0	ular	6	0.0	0.0	0.0
geografiya	4	30.6	33.3	31.2	olomon	3	0.0	0.0	0.0	un	5	0.0	0.0	0.0
gilam	7	17.8	28.6	21.5	olti	4	0.0	0.0	0.0	uning	3	5.6	11.1	7.4
gimnastika	4	56.5	66.7	59.5	olxo'ri	4	38.3	58.3	45.9	unutmoq	3	83.3	77.8	80.0
go'sht	9	14.8	14.8	14.7	ona	4	0.0	0.0	0.0	urishmoq	3	41.1	55.6	44.8
gol	4	0.0	0.0	0.0	opa	3	0.0	0.0	0.0	usta	4	0.0	0.0	0.0
gollandiya	3	0.0	0.0	0.0	oq	7	28.3	23.8	25.6	uy	4	50.0	50.0	49.0
grechka	5	5.6	6.7	6.1	orqada qolmoq	3	22.2	22.2	22.2	uyqusizlik	5	4.8	6.7	5.6
gul	5	0.0	0.0	0.0	oshpaz	4	27.5	25.0	24.1	uzoq	7	41.7	28.6	33.6
guruch	3	0.0	0.0	0.0	oshxona	13	27.2	33.3	29.6	uzum	5	0.0	0.0	0.0
guvoh	4	69.4	58.3	63.1	osmon	5	20.4	26.7	22.4	vagon	3	22.2	22.2	22.2
ha	3	0.0	0.0	0.0	oson	7	15.7	19.0	17.2	vaqt	7	0.0	0.0	0.0
hafta	6	54.6	88.9	67.4	ot	11	27.7	27.3	26.5	vaza	3	63.3	88.9	72.2
hammom	7	5.6	4.8	5.1	ota	5	50.4	80.0	61.1	vazir	3	77.8	66.7	71.1
har-xil	5	0.0	0.0	0.0	ota-ona	4	27.8	16.7	20.6	velosiped	5	0.0	0.0	0.0
harf:a	10	66.1	96.7	78.1	oy	5	0.0	0.0	0.0	vertolyot	5	50.8	80.0	61.2
harf:b	10	50.0	16.7	24.6	oyna	4	0.0	0.0	0.0	vino	5	0.0	0.0	0.0
harf:ch	12	28.2	22.2	24.4	ozarbayjon	3	16.7	11.1	13.3	viza	3	0.0	0.0	0.0
harf:d	10	35.8	40.0	37.4	palto	10	11.1	10.0	10.5	vokzal	5	15.1	20.0	17.2
harf:e	10	0.0	0.0	0.0	panama	8	45.5	45.8	44.8	voleybol	6	38.9	16.7	22.2
harf:f	10	100.0	96.7	98.2	panjara	6	47.6	50.0	48.7	xato	3	0.0	0.0	0.0
harf:g	10	45.9	73.3	55.9	parda	6	45.2	88.9	59.7	xayr	4	40.0	66.7	49.1
harf:g'	10	12.8	16.7	14.5	pardalar	3	0.0	0.0	0.0	xindiston	3	55.6	66.7	56.3
harf:j	9	75.3	100.0	85.8	parket	4	75.0	33.3	43.9	xo'roz	5	6.7	6.7	6.7
harf:k	10	13.9	10.0	11.6	pasport	4	34.7	66.7	45.4	xokkey	5	8.3	6.7	7.4
harf:l	11	0.0	0.0	0.0	paypoq	5	32.8	40.0	35.7	xola	3	0.0	0.0	0.0
harf:m	10	56.7	50.0	53.0	payshanba	5	25.4	33.3	28.8	xona	4	0.0	0.0	0.0
harf:n	10	44.4	13.3	20.2	pechene	6	0.0	0.0	0.0	yakshanba	7	39.4	52.4	44.6
harf:ng	9	81.3	55.6	64.1	pichoq	7	20.1	28.6	23.6	yana	10	41.0	33.3	36.6
harf:o	10	79.1	100.0	88.3	pirog	5	0.0	0.0	0.0	yanvar	7	4.2	4.8	4.4
harf:o'	10	0.0	0.0	0.0	pivo	4	0.0	0.0	0.0	yaponiya	3	44.4	22.2	27.8
harf:p	9	0.0	0.0	0.0	piyodalar o'tish yo'li	5	0.0	0.0	0.0	yashamoq	4	31.7	25.0	26.9
harf:r	10	90.6	90.0	90.2	piyola	5	13.3	13.3	13.3	yashil	4	37.3	58.3	44.9
harf:s	10	70.1	90.0	78.6	piyoz	7	43.6	81.0	56.7	yaxshi	12	5.1	5.6	5.2
harf:sh	12	60.5	69.4	64.0	pochta	3	0.0	0.0	0.0	yemoq	7	39.3	66.7	49.4
harf:t	8	16.7	8.3	11.1	pol	4	62.2	33.3	39.8	yengil	4	0.0	0.0	0.0
harf:u	10	40.2	30.0	30.5	pomidor	4	0.0	0.0	0.0	yer	5	0.0	0.0	0.0
harf:x	10	57.6	73.3	64.3	pora	3	0.0	0.0	0.0	yer osti yo'li	5	0.0	0.0	0.0
harf:z	10	75.9	86.7	80.7	poshna	3	0.0	0.0	0.0	yetmish	3	0.0	0.0	0.0
harf:ж	4	0.0	0.0	0.0	prokuror	7	22.6	19.0	20.3	yetti	4	0.0	0.0	0.0
harf:к	3	0.0	0.0	0.0	pul	4	0.0	0.0	0.0	yiqilmoq	5	23.3	13.3	16.2
harf:н	3	0.0	0.0	0.0	pushti	6	23.3	11.1	14.4	yo'lbars	5	56.5	93.3	70.3
harf:c	3	0.0	0.0	0.0	qabziyat	4	25.0	25.0	25.0	yo'q	12	0.0	0.0	0.0
harf:ф	3	0.0	0.0	0.0	qachon	10	71.9	70.0	70.5	yod olmoq	4	33.3	8.3	13.3
harf:ц	3	23.3	44.4	30.6	qadah	3	0.0	0.0	0.0	yomg'ir	5	6.7	6.7	6.7
harf:ъ	3	0.0	0.0	0.0	qalam	13	0.0	0.0	0.0	yomon	4	0.0	0.0	0.0
harf:ь	4	13.9	16.7	15.0	qalampir	3	0.0	0.0	0.0	yonalish	4	0.0	0.0	0.0
harf:я	3	0.0	0.0	0.0	qamoq	3	6.7	11.1	8.3	yoqadi	4	18.8	41.7	25.6
haydovchi	3	0.0	0.0	0.0	qamoqxona	3	0.0	0.0	0.0	yosh	3	0.0	0.0	0.0
hech kim	4	21.6	50.0	30.0	qancha	3	0.0	0.0	0.0	yostiq	4	10.8	16.7	13.0
hech qachon	4	0.0	0.0	0.0	qandil	6	50.0	50.0	49.4	yovvoyi hayvonlar	4	0.0	0.0	0.0
hikoya	3	16.7	11.1	13.3	qari	3	0.0	0.0	0.0	yoz	9	57.4	55.6	56.4
hojatxona	8	57.8	75.0	64.8	qarindosh	4	35.6	41.7	37.7	yugoslaviya	3	0.0	0.0	0.0
hujjat	4	37.1	58.3	45.1	qaror	3	38.3	55.6	45.2	yugurish	6	17.5	16.7	16.9
hujum	3	0.0	0.0	0.0	qarz	4	0.0	0.0	0.0	yugurmoq	6	4.2	5.6	4.8
hukm	3	0.0	0.0	0.0	qattiqqo'l	4	0.0	0.0	0.0	yuk tashuvchi	4	33.3	33.3	32.9
id	3	0.0	0.0	0.0	qavat	8	18.6	16.7	17.4	yulduz	6	24.4	22.2	23.2
ijara	3	100.0	77.8	86.7	qayerda	13	4.5	5.1	4.8	yupqa	4	0.0	0.0	0.0
ikki	6	0.0	0.0	0.0	qayerga	6	10.5	22.2	13.9	yuqumli kasallik	3	0.0	0.0	0.0
ikki yildan keyin	4	15.0	16.7	15.7	qayiq	9	0.0	0.0	0.0	yurak	4	48.9	50.0	48.7
ilan	6	0.0	0.0	0.0	qaynatmoq	3	0.0	0.0	0.0	yurist	3	0.0	0.0	0.0
iliq	3	0.0	0.0	0.0	qilichbozlik	7	0.0	0.0	0.0	yurmoq	4	0.0	0.0	0.0
ilon	5	0.0	0.0	0.0	qish	14	13.3	11.9	12.5	yutmoq	3	0.0	0.0	0.0
iltimos	11	70.6	57.6	63.3	qisqa	5	17.9	20.0	18.5	yutqazmoq	3	0.0	0.0	0.0
imtiyoz	7	25.9	19.0	21.4	qiyin	5	30.6	33.3	31.6	yuz	3	3.7	11.1	5.6
indinga	3	0.0	0.0	0.0	qiz	5	0.0	0.0	0.0	zaharlanish	3	0.0	0.0	0.0
infarkt	5	40.1	53.3	45.4	qiz bola	5	0.0	0.0	0.0	zaif	3	0.0	0.0	0.0
inson	3	0.0	0.0	0.0	qizil	8	5.7	12.5	7.8	zavod	3	0.0	0.0	0.0
ish	3	0.0	0.0	0.0	qo'lqop	7	35.4	52.4	41.9	zina	7	0.0	0.0	0.0
ishchi	3	51.0	100.0	67.2	qo'rqmoq	6	0.0	0.0	0.0					
ishlab chiqarish	5	4.8	6.7	5.6	qo'shni	3	63.9	66.7	64.1					

S4 Most frequent confusions

Tables S5 and S6 list the most frequent (true class → predicted class) pairs: for Slovo and Bukva under SD-5 (MPE-A1; number of errors per seed, pooled over the five folds), for the UzSL corpus under LOSO (MPE-A1 pretrained on Slovo + Bukva, 682-class head; errors per seed, pooled over the ten folds) and for the UzSL pilot (share of the test recordings of the true class, pooled over the two test participants and three seeds).

Table S5: Most frequent confusions of MPE-A1 under SD-5 on Slovo (left, 20 pairs) and Bukva (right, 20 pairs); errors per seed.

Slovo: true	predicted	errors	Bukva: true	predicted	errors
остановить	остановиться	7.0	Щ	Ш	39.0
наверху	вверх	6.3	Ш	Щ	26.0
краткий	короткий	5.7	М	Т	13.3
остановиться	остановить	5.7	Л	П	13.0
прекрасный	чудесный	5.3	Т	М	12.3
резервирование; бронирование	занятый	5.3	П	Л	11.3
этот	эти	5.3	Ж	Е	7.0
эти	этот	5.0	Е	Ж	6.0
прием пищи	еда	4.7	К	Я	4.7
хотеть	желание	4.7	П	Т	4.7
иметь	Обладать	4.7	Ъ	Ь	3.7
Выигрыш	найти	4.3	Т	П	3.7
коричневый	повар	4.3	Я	Б	3.7
начало	сначала	4.3	И	Ю	3.7
любовь	любовный	4.0	М	Л	3.3
отложенный	отложить	4.0	К	Д	3.3
уверен	уверенный	3.7	Ь	Ъ	3.3
взять	одолжить	3.7	Ж	Ф	3.0
чудесный	прекрасный	3.7	Д	Я	3.0
нога	no_event	3.7	Р	И	2.7

Table S6: Most frequent confusions on the UzSL corpus (left, LOSO, 20 pairs; errors per seed) and on the UzSL pilot (right, 15 pairs; per cent of the test recordings of the true class).

Corpus: true	predicted	errors	Pilot: true	predicted	% of class
harf:g'	harf:g	5.3	0	O	100.0
harf:e	may	5.0	H	X	100.0
harf:u	qayerga	4.7	I	E	100.0
harf:e	harf:x	4.0	V	5	100.0
harf:o'	qayerga	4.0	Y	E	80.0
harf:b	pichoq	3.7	G'	G	73.3
blinchik	quymoq	3.7	9	R	66.7
harf:k	harf:к	3.7	O'	U	66.7
harf:l	mart	3.3	2	D	60.0
likopcha	tarelka	3.3	3	SH	60.0
pechene	yemoq	3.0	5	V	55.6
kreslo	rais	3.0	10	R	53.3
harf:o'	shanba	3.0	4	SH	53.3
ona	ayol	3.0	E	Y	46.7
harf:n	noyabr	3.0	E	X	40.0
shim	sichqon	2.7			
laqab	hujjat	2.7			
milliard	qayerda	2.7			
harf:g	harf:g'	2.7			
harf:n	harf:k	2.7			

S5 Label verification

The label check of Section 3.4 of the article covered 300 recordings (30 per LOSO signer; see Table S7). For 113 of them a reference recording of the same sign existed in the video dictionary (pilot pairs: 57, all judged the same sign; lesson pairs: 56, 54 the same); the reviewer judged 111 of the 113 pairs as the same sign and 2 as different, an accuracy of 98.2% [94.7; 100.0] (percentile bootstrap over signers). For the 187 recordings without a dictionary reference the reviewer compared the recording with the written label alone: 182 judged correct, 3 wrong and 2 uncertain; these judgements are not used as an accuracy estimate.

Table S7: Label check by signer: number of reviewed recordings and the reviewer's judgement (same sign / different sign / uncertain), with and without a dictionary reference combined.

Signer	Reviewed	Same	Different	Uncertain
S01	30	27	3	0
S02	30	30	0	0
S03	30	29	0	1
S04	30	30	0	0
S05	30	30	0	0
S07	30	29	1	0
S08	30	29	0	1
S09	30	29	1	0
S11	30	30	0	0
S12	30	30	0	0
Total	300	293	5	2

S6 Pilot recordings: paired comparisons

Table S8 gives, for each training condition of the pilot, the pooled top-1 accuracy of MPE-A1, of the released SL-GCN and of the graph-free MPE-A2 (all pretrained on Slovo + Bukva where applicable), and the outcome of the pre-registered sign rule: the number of seed–fold pairs (of six) in which the first model is more accurate, less accurate, or equal. A difference is established only when all six pairs have the same sign. Because some test classes do not occur in the training data of a fold, the accuracy reachable on the test participant is bounded by 82.5% (participant 002) and 67.0% (participant 003).

Table S8: UzSL pilot: pooled top-1 accuracy (%) and sign-rule outcomes over the six seed–fold pairs (more / less / equal).

Condition	MPE-A1	SL-GCN	A1 vs SL-GCN	MPE-A2	A2 vs A1
One example per class, fine-tuning	40.3	34.2	5 / 1 / 0	35.4	1 / 5 / 0
One example per class, linear probe	37.2	41.3	1 / 5 / 0	38.0	4 / 2 / 0
One example per class, from scratch	9.5	5.5	5 / 1 / 0	—	—
Two examples per class, fine-tuning	41.2	39.5	3 / 3 / 0	40.5	3 / 2 / 1
Two examples per class, linear probe	39.4	47.2	0 / 6 / 0	40.3	4 / 2 / 0
Two examples per class, from scratch	11.7	5.3	5 / 1 / 0	—	—
All recordings, fine-tuning	47.5	45.9	6 / 0 / 0	45.8	1 / 4 / 1
All recordings, linear probe	45.4	49.8	0 / 5 / 1	47.9	4 / 2 / 0
All recordings, from scratch	24.7	15.2	6 / 0 / 0	—	—

S7 Simulation of transfer on Bukva

Table S9 gives the numbers behind Fig. 7a of the article: MPE-A0 under SD-5 trained from scratch, as a linear probe and with full fine-tuning after pretraining on Slovo, with 1, 3 and 10 training recordings per class and with all training recordings (three seeds; CIs and paired differences over test signers).

Table S9: Bukva under SD-5 with reduced training data (MPE-A0, three seeds): top-1 accuracy [95% CI] and paired differences to training from scratch.

Recordings per class	From scratch	Linear probe (Slovo)	Fine-tuning (Slovo)	Δ probe	Δ fine-tuning
1	10.9 [10.0; 11.9]	70.2 [68.6; 71.7]	70.7 [69.2; 72.1]	+59.2 [+57.2; +61.0]	+59.7 [+57.9; +61.5]
3	26.3 [24.6; 28.0]	81.9 [80.5; 83.2]	81.8 [80.4; 83.0]	+55.6 [+53.6; +57.6]	+55.5 [+53.6; +57.4]
10	63.7 [62.0; 65.4]	86.1 [84.7; 87.3]	86.8 [85.5; 88.0]	+22.4 [+20.6; +24.3]	+23.1 [+21.4; +24.8]
all (about 68)	92.9 [92.0; 93.7]	89.4 [88.4; 90.5]	90.9 [89.9; 91.8]	−3.4 [−4.3; −2.5]	−2.0 [−2.8; −1.2]

S8 Computational cost and exported models

Table S10 gives the measured parameters at inference, floating-point operations per window (`torch.utils.flop_counter`, a multiply–add counted as two operations) and the median CPU latency of 100 runs after 10 warm-up runs (Intel Core i9-14900KF, float32, batch 1, eager PyTorch 2.14.1, default thread count), for the Slovo models at the window used at test time. Latencies of the same model differed by 1–2 ms between sessions. Table S11 gives the TensorFlow Lite exports (float32, tflite-runtime 2.9.1) on the same CPU and the agreement of their argmax with the PyTorch model on the checked recordings.

Table S10: Measured cost of the keypoint models (Slovo) and of the released video model.

Model	Params (M)	GFLOPs / window	CPU (ms)	Frames
MPE-A1	3.38	0.76	9.3	32
MPE, $D = 256$, $N = 6$	10.27	1.33	14.1	32
MPE-A2 (no graph)	3.14	0.196	4.7	32
ST-GCN, re-implemented	3.41	1.98	5.1	32
SL-GCN, re-implemented	3.93	2.03	19.2	32
ST-GCN, released, 32 frames	3.34	1.98	5.7	32
ST-GCN, released, authors' recipe (300 frames)	3.34	18.54	23.2	300
SL-GCN, released, 32 frames	5.07	2.19	23.0	32
SL-GCN, released, authors' recipe (150 frames)	5.07	9.70	26.5	150
MPE-A1, SL-GCN recipe (150 frames)	3.40	3.59	21.2	150
Keypoint Transformer	3.53	0.43	1.8	64
Two-layer Transformer	1.98	0.11	1.7	32
BiLSTM with attention	11.00	0.62	5.2	32
MLP	1.55	0.004	1.0	32
MViTv2-S (RGB video, released)	35.0	—	929.7	32

Table S11: TensorFlow Lite exports: size, median latency per 32-frame window (default threads / one thread) and argmax agreement with PyTorch.

Export	Size (MiB)	Latency (ms)	1 thread (ms)	Checked	Agree (%)
MPE-A1 (Slovo + Bukva heads)	12.5	24.4	29.1	1051	100.0
MPE-A2 (no graph)	11.6	6.6	6.6	1051	100.0
MPE-A1 fine-tuned on the pilot	12.5	24.4	29.1	1051	100.0

Table S12 gives the latency of the ONNX exports in ONNX Runtime 1.31.1 (CPUExecutionProvider, opset 17, batch 1, 32-frame window, trained Slovo weights of fold 1, 10 warm-up and 100 timed runs) with one and four intra-op threads on the same CPU. The measurement was taken while the machine was partly loaded by training jobs (1-minute load average 2.5–4.7 on 32 logical cores), so the values are upper bounds; the maximum absolute difference between ONNX and PyTorch logits is given in the last column.

Table S12: ONNX Runtime latency per 32-frame window: median (P90) in ms with one and four threads.

Model	ONNX (MB)	1 thread	4 threads	max $\lvert\Delta$ logit$\rvert$
MPE-A1	13.0	10.8 (11.4)	7.2 (7.4)	2.4e-06
MPE-A2 (no graph)	12.1	1.8 (2.2)	1.0 (1.1)	4.8e-06
ST-GCN, released	13.0	15.0 (15.2)	7.6 (7.8)	8.6e-06
SL-GCN, released	17.1	25.6 (26.3)	19.3 (19.7)	4.1e-06

S9 Keypoint availability in the UzSL corpus

Table S13 summarises, over the 5,993 recordings of the corpus, the length in frames and the share of frames in which MediaPipe Holistic returned each body part (the detector's presence flags at the default thresholds).

Table S13: Length and keypoint availability of the corpus recordings.

Quantity	Mean	SD	Min	Q1	Median	Q3	Max
Frames per recording	67	50	9	36	52	78	509
Share of frames with pose	1.000	0.000	0.981	1.000	1.000	1.000	1.000
Share of frames with face	0.998	0.020	0.353	1.000	1.000	1.000	1.000
Share of frames with right hand	0.936	0.130	0.000	0.933	1.000	1.000	1.000
Share of frames with left hand	0.777	0.361	0.000	0.762	0.980	1.000	1.000
Share of frames with at least one hand	0.972	0.088	0.062	1.000	1.000	1.000	1.000
Share of frames with both hands	0.741	0.359	0.000	0.638	0.928	1.000	1.000